

Remote Sensing Image Change Caption Generation

Evidence Report — Predictions, Curves, Maps and Plots

Independent Study, Monsoon 2026

Team	Sankalp Chaturvedi (2025201083) KSPSVLN Siddardha Kumar Kavuri (2025201061) Rinkesh Verma (2025201070)
Mentor	Rama Chandra Prasad

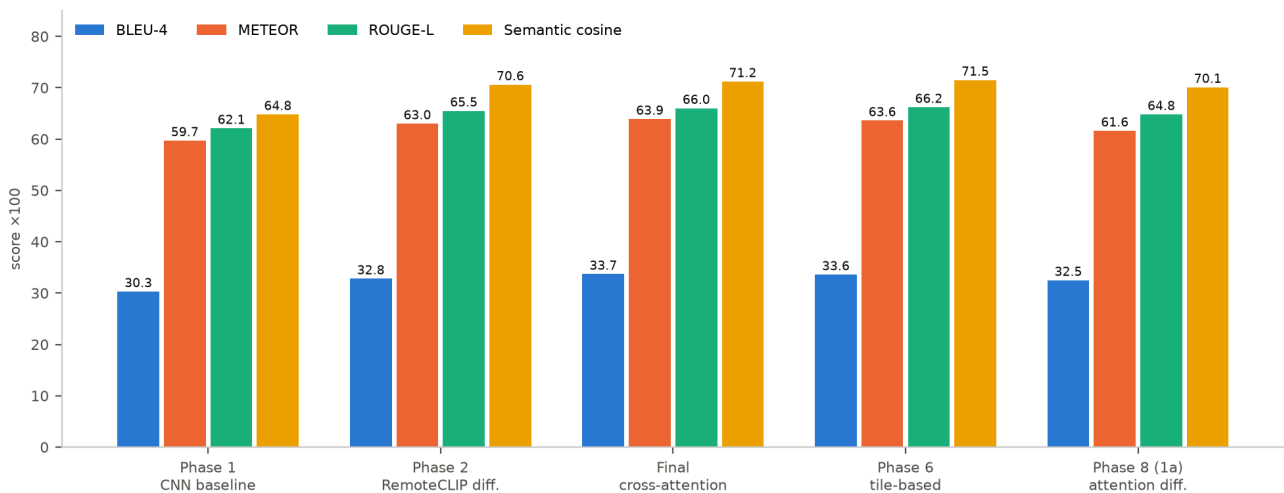
This report collects the evidence behind the observations in the Main Report. Each section names the observation it supports (Obs. 1–15).

How to read this report

- **Sources.**
 - Captioning results come from the notebook outputs of the [project repository](#).
 - Prediction samples come from the saved prediction sheets, published as [RSICC Model Run Reports](#) on Kaggle.
 - The Indian-dataset evidence comes from the experiment logs and the labelled files, published as [India Sentinel-2 Change Pairs](#) and its [labelled subset](#).
- **Ground truth for Part C** is the human label where one exists.
- **Prediction-sheet naming.** The sheets named `predictions_phase7*` contain Phase 6 model outputs. They are labelled "Phase 6 run" here.

Part A — Captioning models on LEVIR-CC and SECOND-CC

E-A1. Scores per model on LEVIR-CC test (Obs. 1, 2)

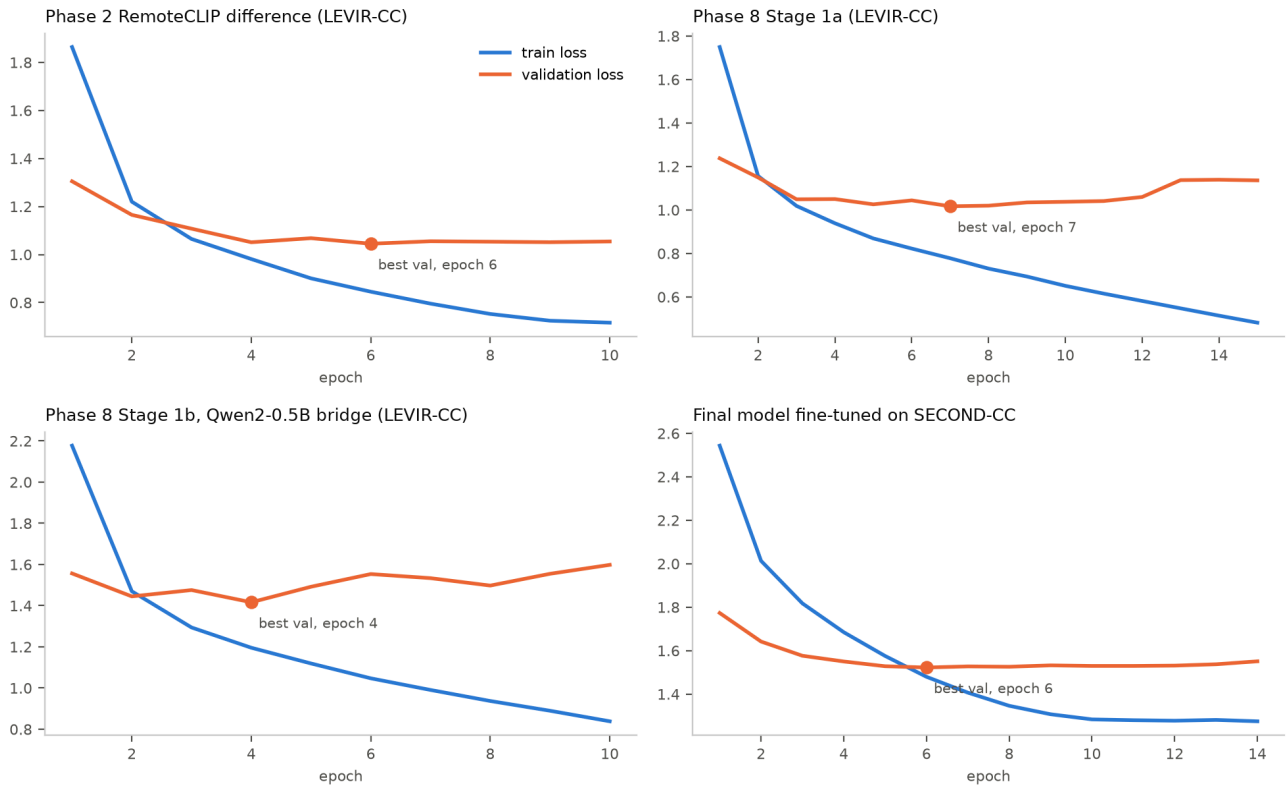


Model	Loss	BLEU-1	BLEU-2	BLEU-3	BLEU-4	METEOR	ROUGE-L	Cosine
Phase 1 baseline	0.9690	0.4720	0.3855	0.3393	0.3027	0.5966	0.6214	0.6477
Phase 2 RemoteCLIP difference	0.9345	0.5168	0.4237	0.3712	0.3284	0.6304	0.6545	0.7061
Final cross-attention	0.8982	0.5311	0.4355	0.3810	0.3365	0.6392	0.6604	0.7123
Phase 6 tiles	0.9225	0.5214	0.4297	0.3777	0.3358	0.6357	0.6624	0.7149
Phase 8 stage 1a (epoch 7)	—	0.5092	0.4167	0.3658	0.3250	0.6155	0.6482	0.7007
Phase 8 stage 1b (Qwen bridge)	—	0	0	0	0	0	0	0.0104

Phase 1 to Phase 2 (encoder only changed, 3.41 M vs 3.43 M trainable parameters):

- BLEU-1 to BLEU-4: +8.5% to +9.9% relative
- METEOR +5.7%, ROUGE-L +5.3%, cosine +9.0%
- test loss −3.6%

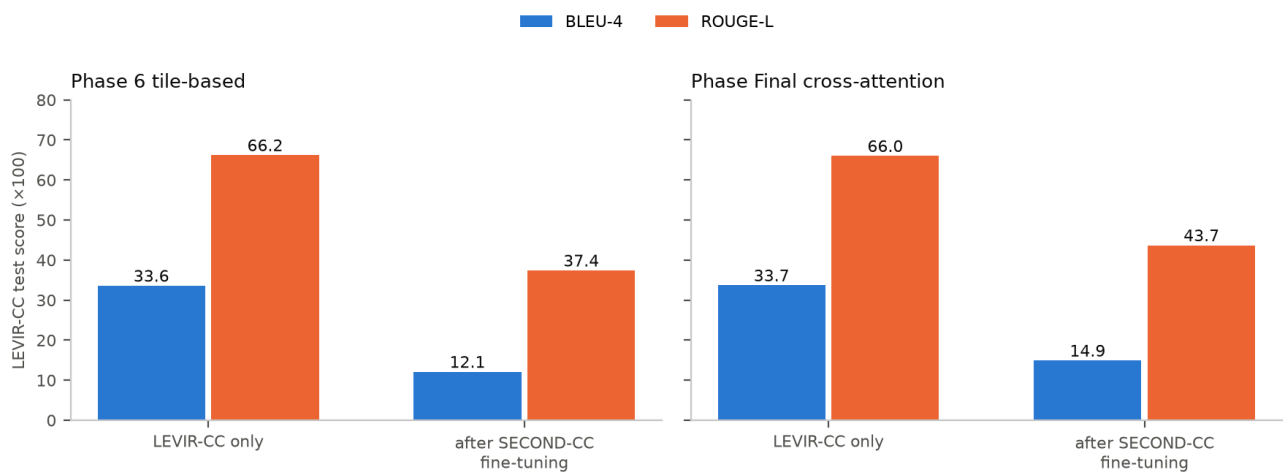
E-A2. Training and validation curves (Obs. 3, 6)



In every run, validation loss reaches its minimum early while training loss keeps falling:

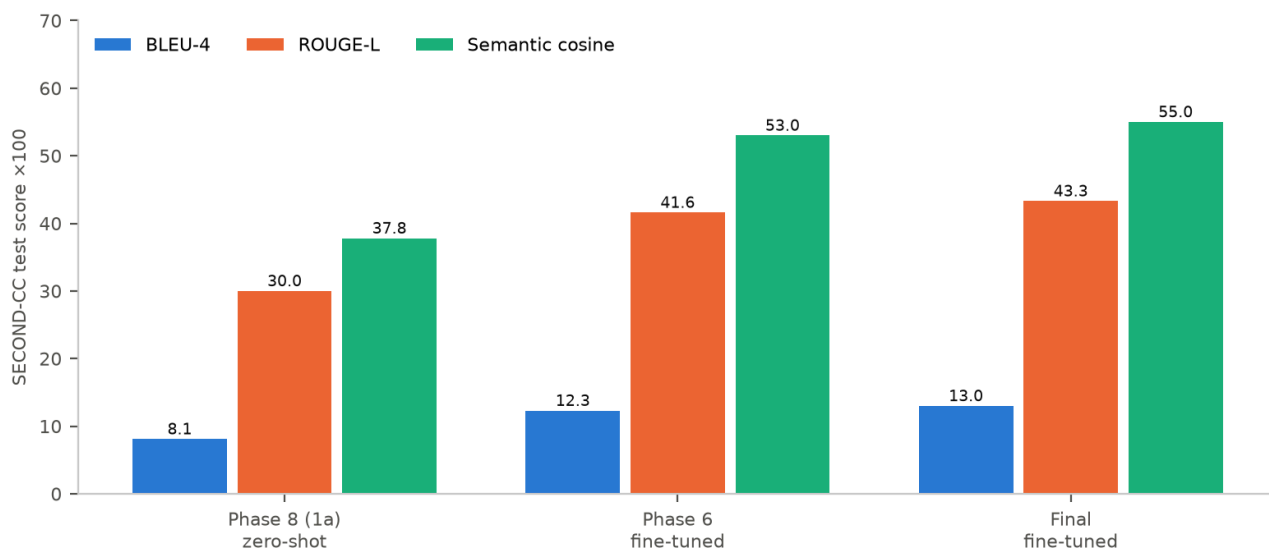
Run	Best validation loss	Final validation loss	Final training loss
Phase 2	1.045 (epoch 6)	1.054 (epoch 10)	0.716
Phase 8, stage 1a	1.017 (epoch 7)	1.136 (epoch 15)	0.482 (training accuracy 84.9%)
Phase 8, stage 1b	1.417 (epoch 4)	1.598 (epoch 10)	—
SECOND-CC fine-tuning	1.522 (epoch 6)	1.551 (epoch 14)	—

E-A3. Forgetting after SECOND-CC fine-tuning (Obs. 6)



Model	LEVIR-CC loss	BLEU-4	METEOR	ROUGE-L	Cosine
Final, before fine-tuning	0.898	0.337	0.639	0.660	0.712
Final, after fine-tuning	4.080	0.149	0.394	0.437	0.458
Phase 6, before fine-tuning	0.923	0.336	0.636	0.662	0.715
Phase 6, after fine-tuning	3.520	0.121	0.318	0.374	0.438

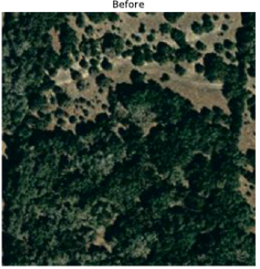
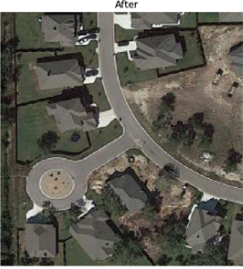
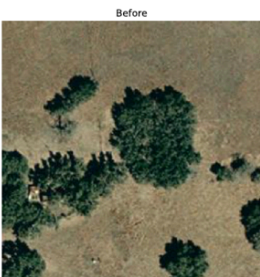
E-A4. Domain shift to SECOND-CC (Obs. 5)



Model / condition	BLEU-1	BLEU-4	METEOR	ROUGE-L	Cosine
Phase 8 stage 1a, zero-shot	0.179	0.081	0.298	0.300	0.378
Phase 6, fine-tuned	0.354	0.123	0.392	0.416	0.530
Final, fine-tuned	0.363	0.130	0.413	0.433	0.550

The zero-shot and fine-tuned rows come from different models, so this comparison is indicative.

E-A5. The same LEVIR-CC test pairs across three runs (Obs. 4, 7)

Before 	After 	LEVIR-CC test sample 4 Reference: the vegetation has been replaced by a road and many villas around . Phase 6 run: a huge residential area is built in almost whole scene Phase 8 run: the vegetation quantity is increased in many parts of the scene Final, after SECOND-CC fine-tuning: there is no difference
Before 	After 	LEVIR-CC test sample 17 Reference: a road appears at the bottom and many houses are scattered replacing the trees . Phase 6 run: a huge residential area is built in almost whole scene Phase 8 run: many buildings are constructed in the middle of the scene Final, after SECOND-CC fine-tuning: there is no difference
Before 	After 	LEVIR-CC test sample 19 Reference: the trees have been replaced by many buildings . Phase 6 run: a huge residential area is built in almost whole scene Phase 8 run: many buildings are constructed in the middle of the scene Final, after SECOND-CC fine-tuning: there is no difference
Before 	After 	LEVIR-CC test sample 42 Reference: a house appears at the top right corner . Phase 6 run: there is no difference Phase 8 run: there is no difference Final, after SECOND-CC fine-tuning: there is no difference
Before 	After 	LEVIR-CC test sample 2 Reference: there is no difference . Phase 6 run: the vegetation quantity is decreased in many parts of the scene Phase 8 run: there is no difference Final, after SECOND-CC fine-tuning: there is no difference

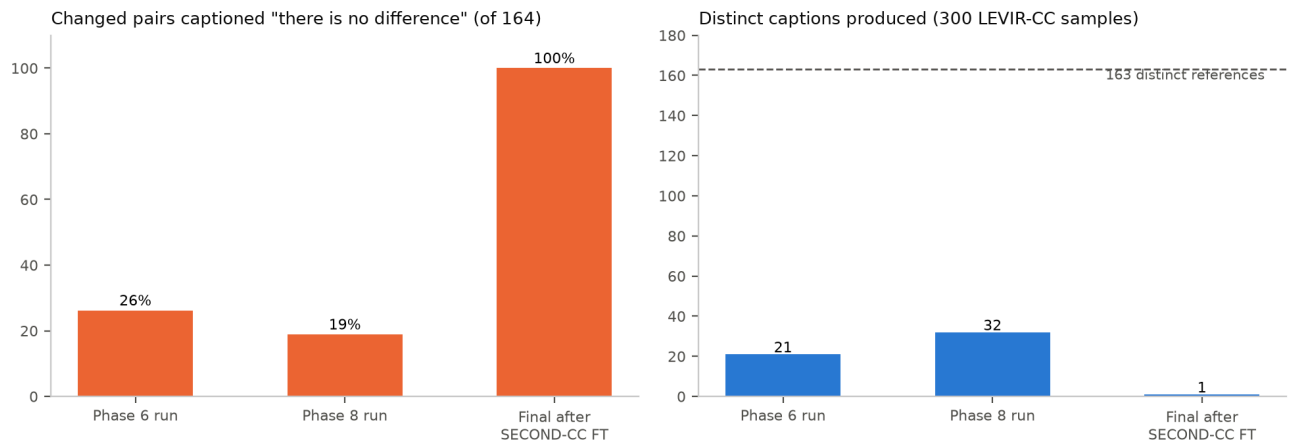
Sample 4, 17 and 19 (forest or scrub replaced by a housing estate):

- The Phase 6 run gives a template ("a huge residential area is built...").
- The Phase 8 run partly contradicts the scene ("vegetation quantity is increased").
- The Final model after fine-tuning says "there is no difference".

Sample 42 (one house added at the top right): all three runs miss it.

Sample 2 (no change): the Phase 6 run reports vegetation loss.

E-A6. Prediction behaviour across 300 saved LEVIR-CC samples (Obs. 4, 7)

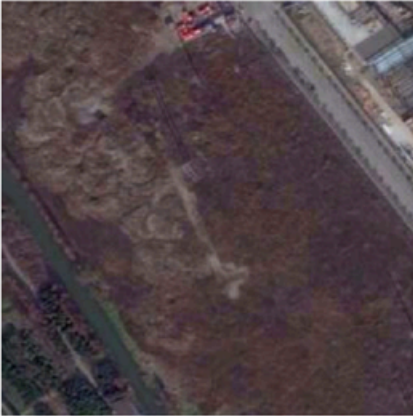
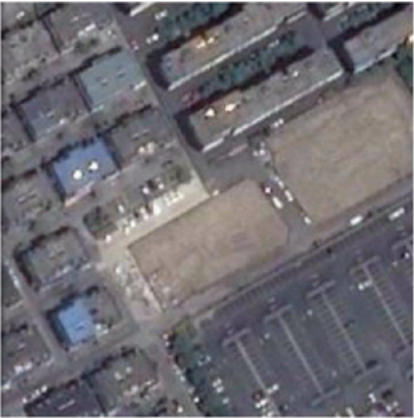
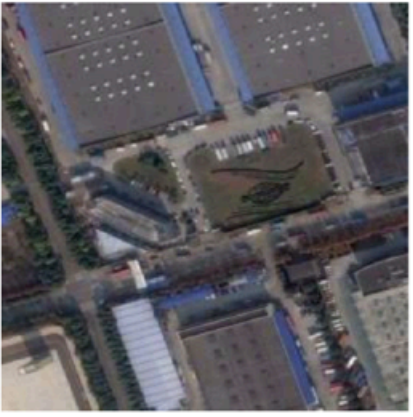


Prediction sheet	Changed pairs called "no difference"	Unchanged pairs correct	Distinct captions
Phase 6 run	43 / 164	98 / 136	21
Phase 8 run	31 / 164	96 / 136	32
Final after SECOND-CC fine-tuning	164 / 164	136 / 136	1
References	—	—	163

On the SECOND-CC prediction sheets (300 samples, 255 changed), the Phase 6 and Phase 8 runs produced 97 and 82 distinct captions. They missed 16 and 19 changed pairs, and called 12 and 15 of 45 unchanged pairs changed.

E-A7. SECOND-CC test predictions (Obs. 5)

SECOND-CC test samples (Phase 6 run)

Before 	After 	Sample: 1 Reference: the trees is turned into bareland at the middle of the scene Prediction: the vegetation quantity is increased in the middle of the scene
Before 	After 	Sample: 5 Reference: trees are removed Prediction: the vegetation quantity is increased in the middle of the scene
Before 	After 	Sample: 9 Reference: lush area turned bareland in the center Prediction: the blue building at the corner of the scene is replaced with a dirt area

E-A8. Whole-image vs patch-wise inference on unlabelled high-resolution pairs (Obs. 9)

Whole-image inference



Patch-wise inference



These 43 pairs have no reference captions. In the samples shown, the after images contain new farm structures and buildings.

- **Whole-image inference** (23 distinct captions) mostly reports "vegetation quantity is decreased".
- **Patch-wise inference** (9 distinct captions) gives "the green region is replaced by a huge house at the bottom right" for 75% of pairs, regardless of where the change is.

Part B — Off-the-shelf captioners on Indian imagery

E-B1. Pilot Indian Sentinel-2 collection



60 areas of interest in 5 categories, tiled into 512×512 patches and filtered by change magnitude. Of 2,126 tiles checked, 1,072 pairs were kept, in 1 h 53 min with 12 workers.

E-B2. Captioner outputs (Obs. 10)



Before

After

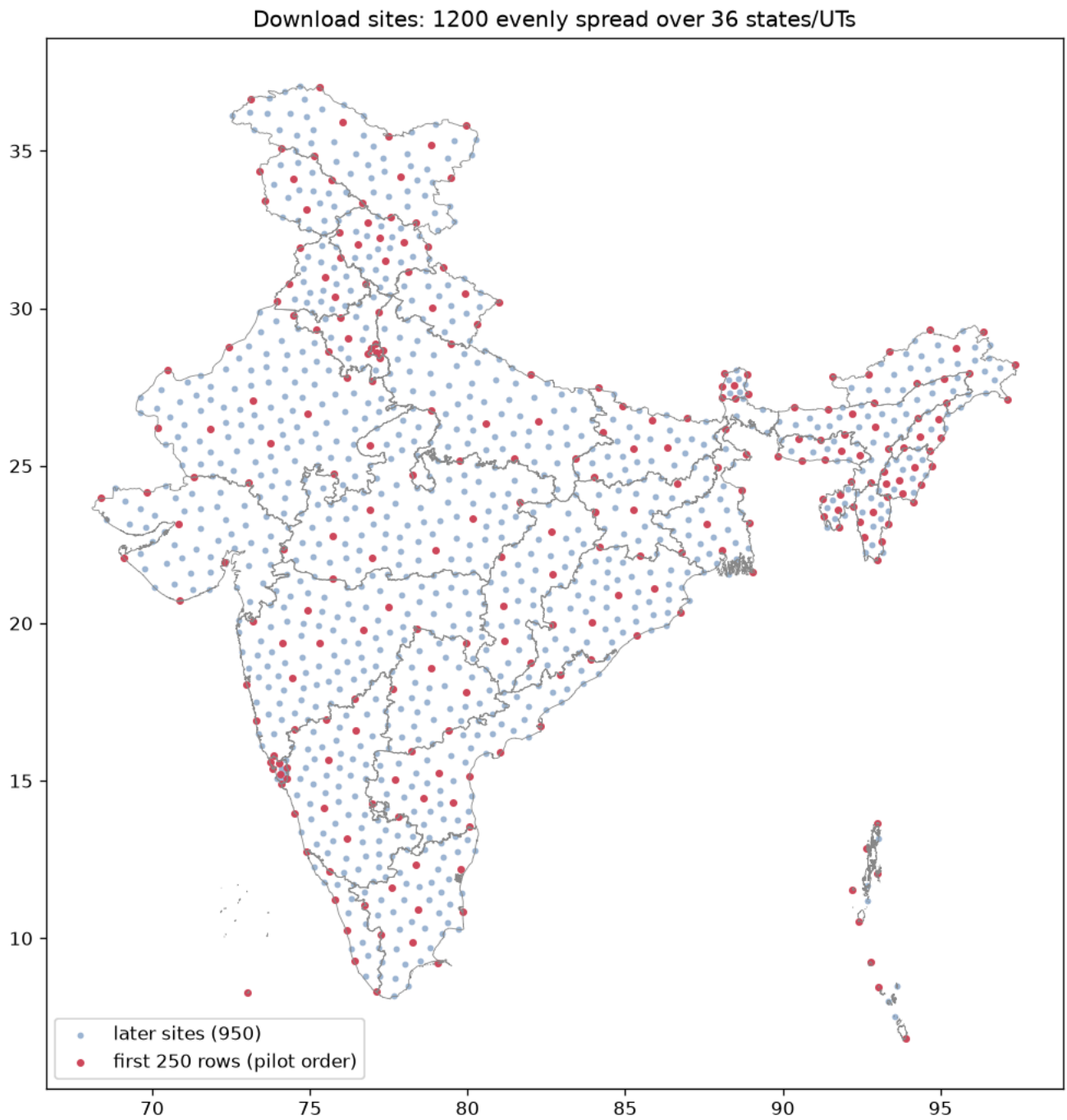
Qwen2.5-VL: "...increased urban density and expansion of buildings, particularly along the central axis, indicating rapid development. There's also visible growth in road infrastructure..."

TeoChat: "Nothing meaningful changed."

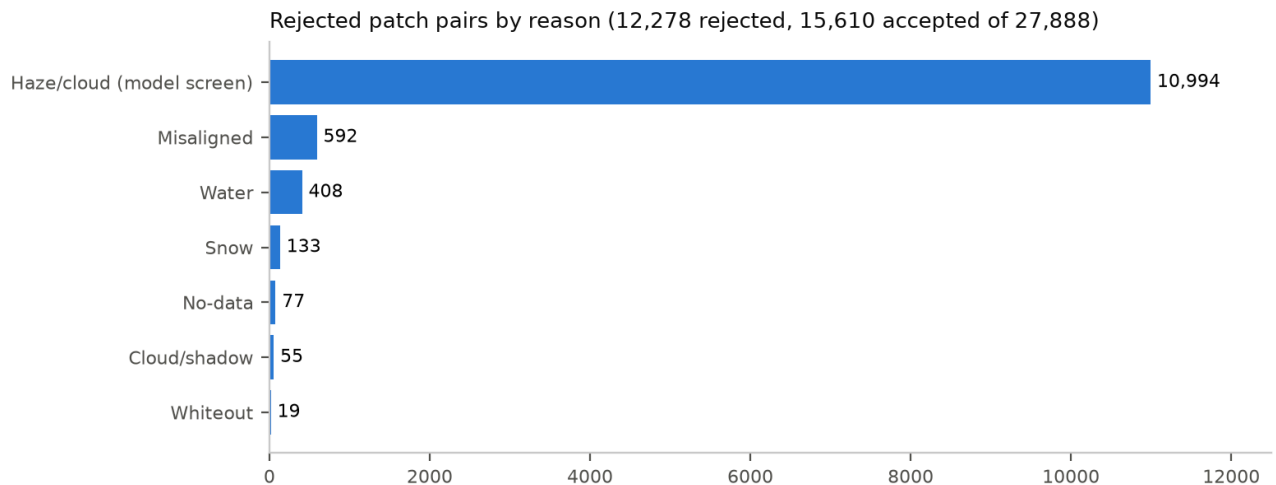
Captioner (30 pairs)	Outcome
TeoChat	"Nothing meaningful changed" on 27/30; one invented hurricane-damage description
TeoChat, 4-bit	"Nothing meaningful changed" on 23/30
Qwen2.5-VL, first run	11 empty and 12 garbled answers (non-English text, repeated tokens)
Qwen2.5-VL, retries / 8-bit	Failures reduced but not removed; answers describe generic urban development

Part C — India-wide dataset and automatic labelling

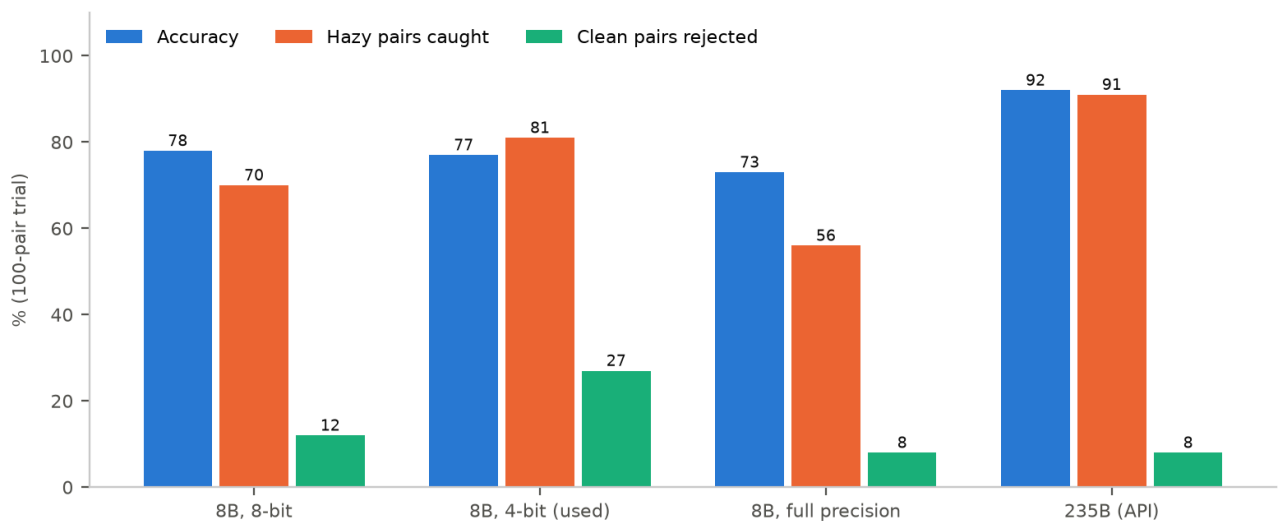
E-C1. Site coverage



E-C2. Screening outcome (Obs. 11)



E-C3. Haze screen trial, 100 pairs (Obs. 11)



The 4-bit model was used for the full run because it was 8× faster on the cluster GPU (1.4 s per pair). The cost was a 27% wrong-reject rate on clean pairs. The reference labels for this trial were re-checked at large size using the 235B model as a second opinion, which favours the 235B row; its accuracy should be read as an upper bound.

E-C4. Label classes and human review (Obs. 12)

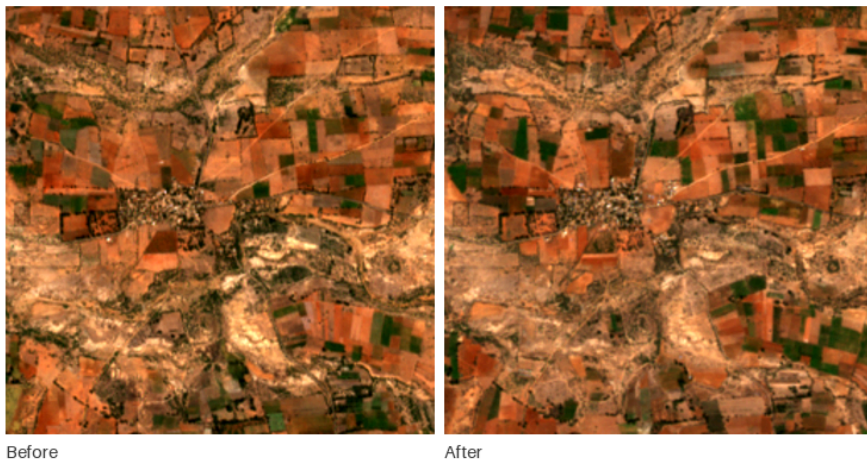
Human label: Change (andhra_pradesh_t1_024_r1c1)



Human label: Seasonal (andhra_pradesh_001_r0c2)



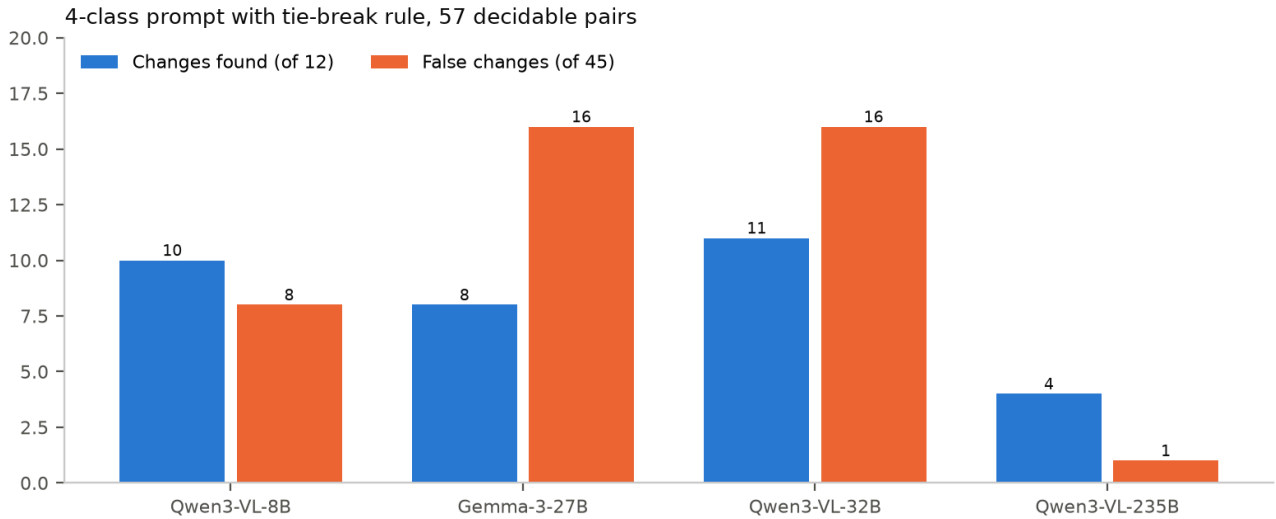
Human label: No change (andhra_pradesh_001_r1c2)



	Change	Seasonal	No change	Can't tell	Total
Human-labelled (blind)	55	214	226	40	535
All labelled pairs	143	400	383	76	1,002

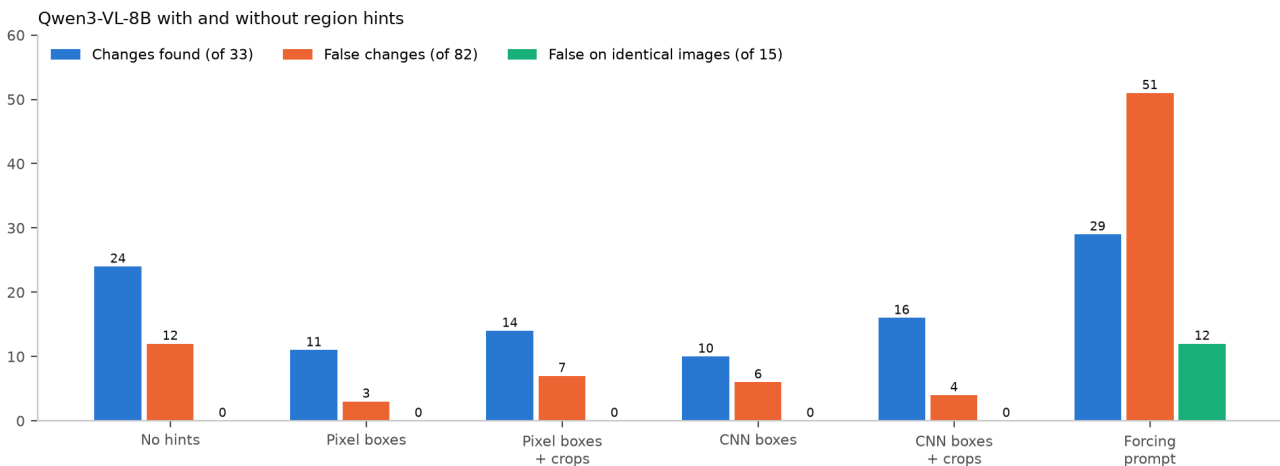
- The human agreed with the automatic suggestion on 84% of reviewed pairs.
- In a random sample of 45 pairs where everything agreed, the human disagreed on 9: 4 changes everyone missed, and 5 false changes everyone accepted.

E-C5. Model size comparison (Obs. 13)



Asked plainly, the 235B dismissed real construction as "seasonal or agricultural variation" in most missed cases. The 32B over-called change.

E-C6. Effect of region hints and forcing prompts (Obs. 14)



The no-hint baseline used the 4-class prompt, while the boxed variants used a per-box prompt; the box variants are directly comparable with each other. With region boxes, the model's overall answer matched its answer for the boxes. It never reported a change outside the boxes, even when asked to.

E-C7. Example: region boxes hide a real change (Obs. 14)

Plain prompt: change found (correct). Human label: Change (bihar_004_r2c1)



Before



After

Pixel boxes: answer 'none' (missed)

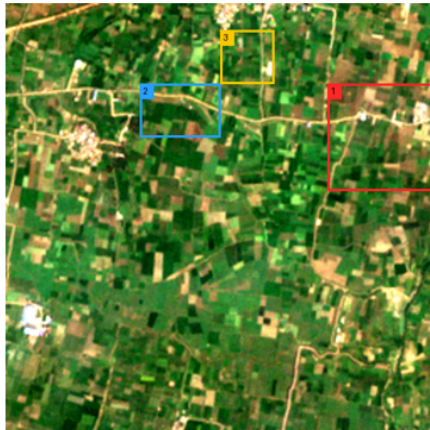


Before, pixel-difference boxes



After, pixel-difference boxes

CNN boxes: answer 'seasonal' (missed)



Before, CNN boxes

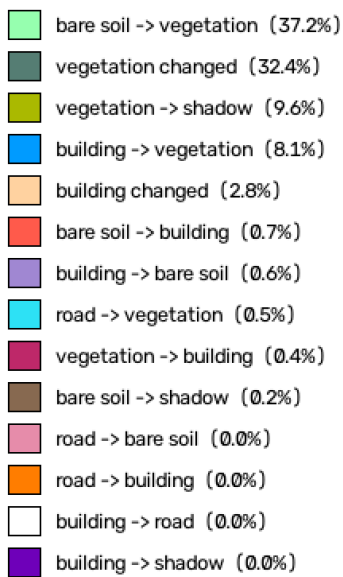


After, CNN boxes

E-C8. Rule-based change map (Obs. 14)

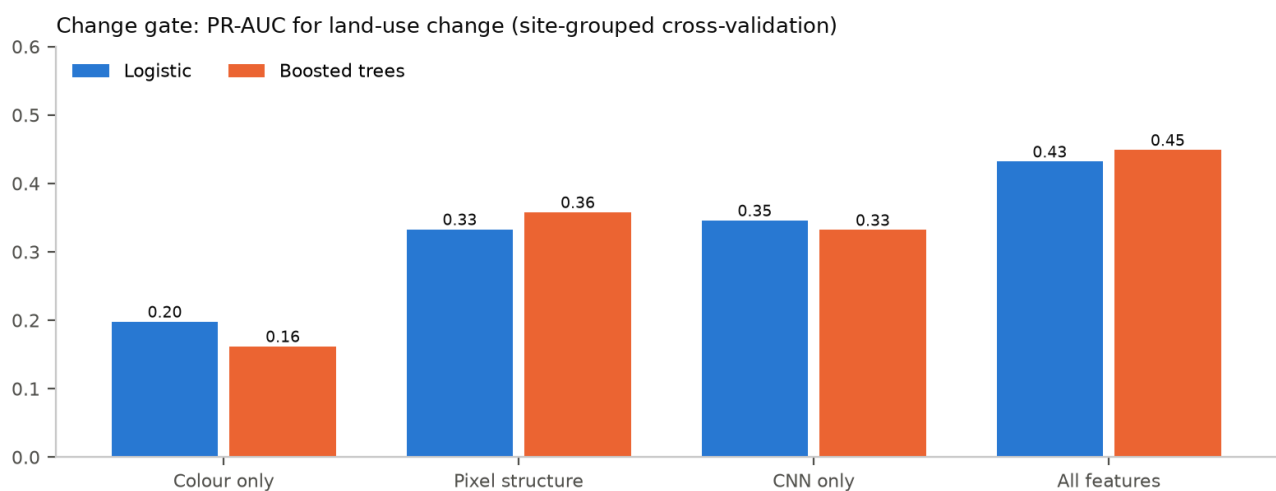
Human label: Seasonal (madhya_pradesh_071_r3c3). Before | After | change map

32B caption with map: "Large areas of bare soil became vegetated, vegetation changed across the scene, and buildings were removed in the left and bottom-left, leaving vegetation."



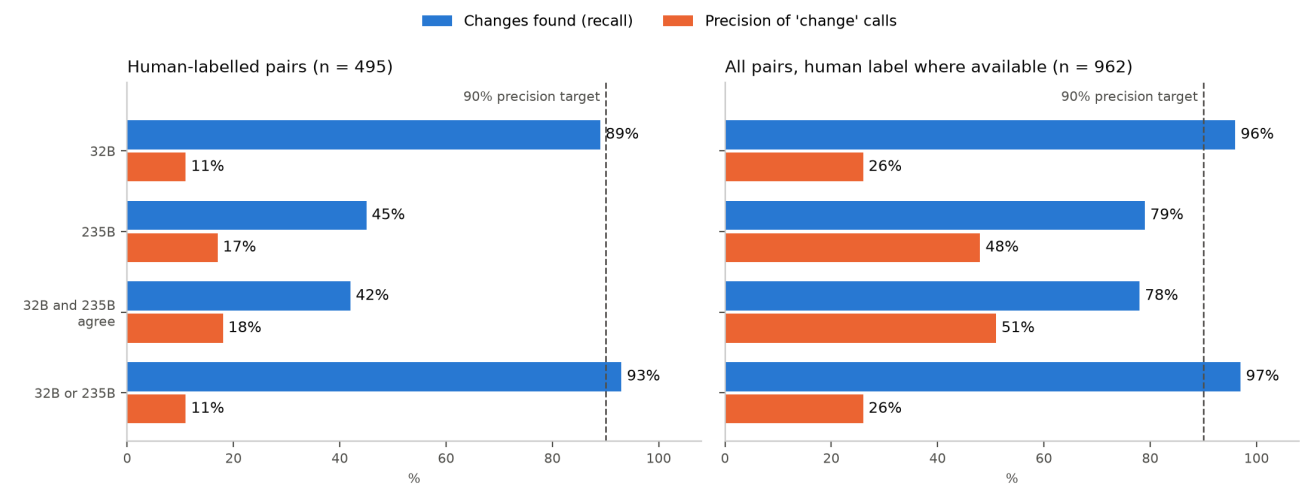
On 50 pairs, the map marked a median 82% of each image as changed (maximum 98%). With the map treated as fact, the model reported change on 19/19 real-change pairs and on 27/27 no-change pairs.

E-C9. Pixel and CNN feature change gate



At 90% recall the gate keeps about 62% of pairs, so it can only rank pairs, not label them.

E-C10. 1,000-pair validation (Obs. 15)



Human-labelled set (n = 495), with 95% Wilson intervals:

Rule	Changes found	Precision	False-change rate
32B	89% (78–95)	11% (9–15)	87% (84–90)
235B	45% (33–58)	17% (12–24)	27% (23–32)
Both agree	42% (30–55)	18% (12–25)	24% (21–29)

How often each model said "change", by human label:

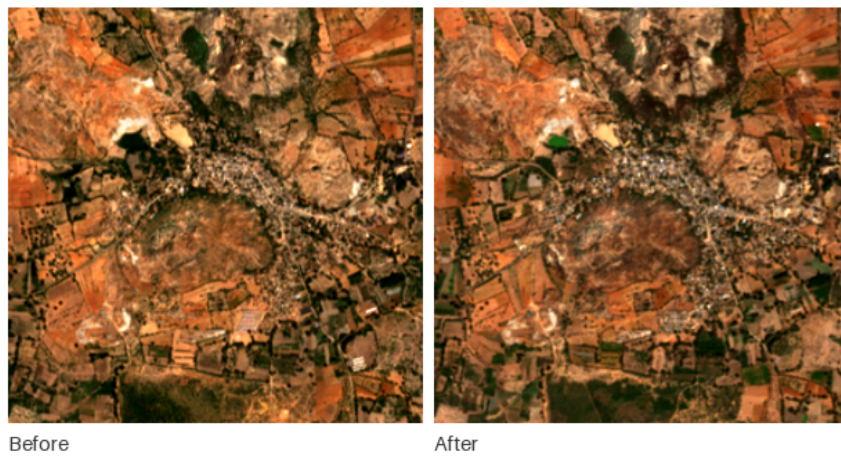
Human label	32B says change	235B says change
Seasonal (n = 214)	200 (93%)	67 (31%)
No change (n = 226)	184 (81%)	53 (23%)
Change (n = 55)	49 (89%)	25 (45%)

E-C11. Validation examples (Obs. 13, 15)

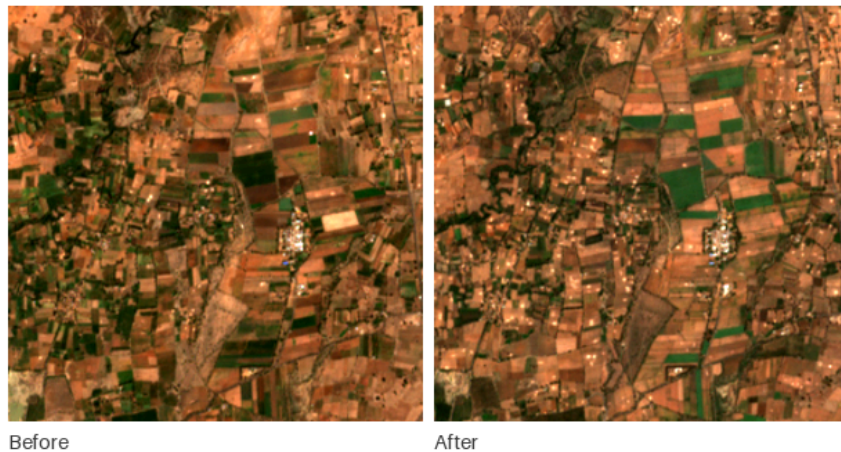
(a) Correct: human Change; 32B and 235B say change



(b) False alarm: human Seasonal; 32B and 235B say change



(c) Missed: human Change; 32B and 235B say seasonal



The 235B's conclusions in its own words:

- **(a) Correct, human label Change:** *"There has been significant, lasting development, including new buildings, roads, and land clearing."*
 - **(b) False alarm, human label Seasonal:** *"Lasting changes have occurred, including new agricultural plots, possible expansion of settlement."*
 - **(c) Missed, human label Change:** *"The differences are not lasting structural changes. They are consistent with seasonal or annual agricultural cycles."*
-

Resources

Resource	Link
Code repository	github.com/Sankalp0109/Generative-and-Caption-Based-Vision-Language-Models-for-Remote-Sensing-Change-Detection
India Sentinel-2 Change Pairs (15,610 pairs)	kaggle.com/datasets/kspsvlnsiddardha/india-sentinel2-change-pairs
Labelled subset (1,002 pairs)	kaggle.com/datasets/kspsvlnsiddardha/india-sentinel2-change-labelled
Model run reports (prediction sheets)	kaggle.com/datasets/kspsvlnsiddardha/rsicc-model-run-reports
Model checkpoints	kaggle.com/models/kspsvlnsiddardha/rsicc-change-captioning · huggingface.co/Kspsvln/IS · huggingface.co/Kspsvln/IS_phase2

Datasets, models and metrics are cited in the References of the Main Report. Contains modified Copernicus Sentinel data (2019–2026).